# The f(R) = R$^{(}67/48)Gravitational Action and Horndeski EFT$

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**Abstract.** We formulate the gravitational action of SDGFT as an f(R) = R$^{(}67/48) modified gravity theor$
$Sawicki Horndeski parameters, showing that the Planck mass run-rate is alpha_M =$
$19/86, the braiding is alpha_B = -19/172, and the tensor speed excess is alpha_T = 0, in compliance with the GV$

## 1 Introduction

Modified gravity theories often introduce arbitrary functions. In SDGFT, the f(R) action is fixed by the effective spectral dimension.

## 2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g}\, R^{67/48} \tag{1}$$

$$\alpha_{\mathrm{M}} = \frac{n-1}{2n-1} = \frac{19}{86} \approx 0.2209, \quad \alpha_{\mathrm{B}} = -\frac{\alpha_{\mathrm{M}}}{2}, \quad \alpha_{\mathrm{T}} = 0 \tag{2}$$

## 3 Horndeski Parameterization

We compute the Bellini-Sawicki parameters for f(R) = R$^{n} and show that the gravitational wave propagation speed i$

## 4 Conclusion

The f(R) = R$^{(}67/48) theory is stable and matches all solar system and gravitational wave constraints.$

## References

[1] D. A. Besemer, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).

[2] D. A. Besemer, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04.

[3] D. A. Besemer, `sdgft` Python package, https://github.com/david-besemer/sdgft (2026).